

Anton Izosimov – Curriculum Vitae

Reader

School of Mathematics and Statistics
University of Glasgow

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EDUCATION

- Loughborough University, PhD in Mathematics, 2012
Thesis title: Singularities of bi-Hamiltonian systems and the multidimensional rigid body
Advisor: Alexey Bolsinov
- Moscow State University, Candidate of Sciences in Mathematics, 2011
Thesis title: Focus-focus singularities of integrable Hamiltonian systems
Advisors: Alexey Bolsinov, Andrej Oshemkov
- Moscow State University, MS with honors in Mathematics, 2008

EMPLOYMENT

- University of Glasgow, Reader, 2025 - present
- University of Arizona, Assistant and Associate Professor, 2018 - 2025
- University of Toronto, Postdoctoral Fellow, 2014 - 2017
- National Research University Higher School of Economics, Docent, 2013 - 2014
- Moscow State University, Assistant Professor, 2012 - 2014

VISITING POSITIONS

- Max Planck Institute for Mathematics, January - August 2025
- Max Planck Institute for Mathematics, June - December 2017
- Hausdorff Research Institute For Mathematics, January - April 2012

TEACHING EXPERIENCE

- University of Glasgow: Math 1 (Core Skills), Introduction To Real Analysis, Galois Theory
- University of Arizona
 - Undergraduate: Calculus I and II, Formal Mathematical Reasoning and Writing, Introduction to Linear Algebra, ODEs and Stability Theory, Real Analysis, Theory of Complex Variables, Theory of Probability, Theory of Statistics
 - Graduate: Algebra, Complex Analysis, Geometry and Topology, Symplectic Geometry and Integrable Systems, Theory of Graphs and Networks
- University of Toronto: Calculus, Differential Equations, Groups and Symmetry, Linear Algebra, Real Analysis
- Higher School of Economics: Calculus, Linear Algebra
- Moscow State University: Geometry and Topology

SELECTED AWARDS, GRANTS, AND SCHOLARSHIPS

- Simons Foundation Travel Support for Mathematicians, 2024 - 2025
- NSF Research Grant, 2020 - 2024
- F.V. Atkinson Teaching Award, University of Toronto, 2016
- Dynasty Foundation Scholarship, Russia, 2013 - 2014

SERVICE

To the profession and broader community:

- Grant reviewer/panelist for NSF and the Simons Foundation
- Referee for several book series and multiple journals, including *Inventiones*, *Duke*, etc.
- Advisory committee member, *Poisson* conference series
- Guest editor of the *Arnold Mathematical Journal* (special issue on finite dimensional integrable systems), 2025
- Co-organizer of the *Lie groups, geometry, integrability, hydrodynamics* conference, Nordfjordeid, Norway, 2024
- Co-organizer of the virtual Hamiltonian Systems Seminar, 2020 - present

To the University of Glasgow:

- London Mathematical Society representative
- Co-organizer of the Integrable Systems and Mathematical Physics Seminar, 2025 - present
- Co-organizer of the joint Edinburgh-Glasgow Integrability Seminar, 2025 - present

To the University of Arizona:

- Committee membership: Graduate (2020 - 2022), Postdoctoral (2020 and 2022 - 2024), Personnel (2024 - 2025)
- Coordinator of the Undergraduate Research Assistantship Program, 2020 - 2021
- Co-chair of the Mathematics Colloquium, 2018 - 2019

To the University of Toronto:

- Organizer of the Symplectic Geometry Seminar, 2015 - 2016

GRADUATE STUDENTS

- Quinton Aboud, University of Arizona, PhD, 2026
- Abigayle Dirdak, University of Arizona, PhD, 2024
- Ilia Kirillov, University of Toronto, PhD, 2024 (co-advised with B. Khesin)
- Leah Hand, University of Arizona, PhD, 2024
- Quinton Aboud, University of Arizona, MS, 2021
- Abigayle Dirdak, University of Arizona, MS, 2021
- Ilia Kirillov, Moscow State University, MS, 2019
- Daniel Fusca, University of Toronto, PhD, 2018 (co-advised with B. Khesin)
- Konstantin Aleshkin, Moscow State University, MS, 2015

POSTDOCS

- Darlayne Addabbo, University of Arizona, 2020 - 2024 (now a TTAP at SUNY Poly)
- Melinda Lanius, University of Arizona, 2018 - 2021 (now a TTAP at Auburn)

INVITED CONFERENCE TALKS AND LECTURE SERIES

Conferences:

- Poisson 2026, Leuven, Belgium, 2026
- Cluster Algebras and Mathematical Physics in the Lakes, University of Cumbria, 2025
- Statistical Mechanics and Discrete Geometry, IPAM, Los Angeles, 2024
- Infinite Dimensional Geometry and PDEs, Banff, 2023
- Math in the Mine: a workshop on shape spaces, Minière de Vallauria, France, 2023
- Finite Dimensional Integrability in Mathematical Physics, Les Diablerets, 2023
- JMM, Special Session on Integrable Systems and Symplectic Group Actions, Boston, 2023
- FDIS 2022: Finite Dimensional Integrable Systems, Tel Aviv University, 2022
- AMS Fall Eastern Sectional Meeting, Session on Geometric Dynamics, Online, 2020
- FDIS 2019: Finite Dimensional Integrable Systems, Shanghai Jiao Tong University, 2019
- Cluster Algebra and Mathematical Physics, Michigan State University, 2018

- Geometric Aspects of Momentum Maps and Integrability, CSF Ascona, 2018
- Gone Fishing 2018: a meeting on Poisson geometry, UC San Diego, 2018
- Analytic and Algebraic Methods in Differential Equations, Moscow, 2017
- Painleve Equations and Discrete Dynamics, Banff, 2016
- Integrable Systems, CSF Ascona, Switzerland, 2016
- Integrability and Near-Integrability in Mechanics and Geometry, BIRS-Oaxaca, 2016
- Analysis of PDEs of Fluid Mechanics, Rice University, 2016
- Integrability in Mechanics and Geometry, ICERM, 2015
- Beyond Toric Integrability, EPFL, 2013

One-day workshops:

- Clusters and their Applications in the North VI, University of Edinburgh, 2026
- Mathematical Aspects of Integrable Systems, Heriot-Watt University, 2026

Lecture series:

- Infinite-dimensional Geometry and Integrable PDEs, *Infinite-dimensional Geometry: Theory and Applications* Thematic Program, Erwin Schrödinger Institute, 2025
- Poisson Geometry and Hydrodynamics, *Poisson* Summer School, Fields Institute, 2018

PAPERS

Preprints:

1. Integrability of Cauchy problems for discrete conformal maps and circle patterns, with M. Arnold, arXiv:2607.08901
2. Virasoro extensions for diffeomorphisms with breaks, with B. Khesin and H. Xiao, arXiv:2605.07682
3. Surface topology and incidence theorems over division rings, arXiv:2603.00288
4. The dimer model and dynamical incidence geometry, with P. Pylyavskyy, arXiv:2512.15499

Papers:

1. Pentagon maps over rings, Grassmannians, and skewers, with L. Hand
Journal of the European Mathematical Society, 2026
2. Folding of quadrilaterals, zigzags, and Arnold-Liouville integrability
Arnold Mathematical Journal, vol. 11, 2025
3. Planar networks and simple Lie groups beyond type A
Advances in Mathematics, vol. 480, p. 110482, 2025
4. Coadjoint orbits of area-preserving diffeomorphisms of non-orientable surfaces, with B. Khesin and I. Kirillov
Journal of Symplectic Geometry, vol. 23, no. 1, pp. 1-35, 2025
5. Integrable systems and cluster algebras, with M. Gekhtman
Encyclopedia of Mathematical Physics 2nd edition, 2024
6. Geometry of generalized fluid flows, with B. Khesin
Calculus of Variations and Partial Differential Equations, vol. 63, no. 3, 2024
7. Change of polytope volumes under Möbius transformations and the circumcenter of mass
Discrete & Computational Geometry, vol. 72, pp. 1369-1376, 2024
8. Long-diagonal pentagon maps, with B. Khesin
Bulletin of the London Mathematical Society, vol. 55, no. 3, pp. 1314-1329, 2023
9. Polygon recutting as a cluster integrable system
Selecta Mathematica, vol. 29, no. 21, 2023
10. Jordan-Kronecker invariants of Lie algebra representations and degrees of invariant polynomials, with A. Bolsinov and I. Kozlov
Transformation Groups, vol. 28, pp. 541-560, 2023
11. What is a lattice W-algebra, with G. Marí Beffa
International Mathematics Research Notices, no. 19, pp. 17021-17059, 2023

12. Dimers, networks, and cluster integrable systems
Geometric and Functional Analysis, vol. 32, pp. 861-880, 2022
13. The pentagram map, Poncelet polygons, and commuting difference operators
Compositio Mathematica, vol. 158, no. 5, pp. 1084-1124, 2022
14. Pentagram maps and refactorization in Poisson-Lie groups
Advances in Mathematics, vol. 404, p. 108476, 2022
15. Intersecting the sides of a polygon
Proceedings of the American Mathematical Society, vol. 150, pp. 639-649, 2022
16. The limit point of the pentagram map and infinitesimal monodromy, with Q. Aboud
International Mathematics Research Notices, no. 7, pp. 5383-5397, 2022
17. Smooth invariants of focus-focus singularities and obstructions to product decomposition, with A. Bolsinov
Journal of Symplectic Geometry, vol. 17, no. 6, pp. 1613-1648, 2019
18. Vortex sheets and diffeomorphism groupoids, with B. Khesin
Advances in Mathematics, vol. 338, pp. 447-501, 2018
19. Characterization of steady solutions to the 2D Euler equation, with B. Khesin
International Mathematics Research Notices, no. 24, pp. 7459-7503, 2017
20. Singularities of integrable systems and algebraic curves
International Mathematics Research Notices, no. 18, pp. 5475-5524, 2017
21. Classification of Casimirs in 2D hydrodynamics, with B. Khesin
Moscow Mathematical Journal, vol. 17, pp. 699-716, 2017
22. Finite-dimensional integrable systems: a collection of research problems, with A. Bolsinov and D. Tsonev
Journal of Geometry and Physics, vol. 115, pp. 2-15, 2017
23. Euler equations on the general linear group, cubic curves, and inscribed hexagons, with K. Aleshkin
L'Enseignement Mathématique, vol. 62, pp. 143-170, 2016
24. Coadjoint orbits of symplectic diffeomorphisms of surfaces and ideal hydrodynamics, with B. Khesin and M. Mousavi
Annales de l'Institut Fourier, vol. 66, no. 6, pp. 2385-2433, 2016
25. Pentagrams, inscribed polygons, and Prym varieties
Electronic Research Announcements in Mathematical Sciences, vol. 23, pp. 25-40, 2016
26. Flat bi-Hamiltonian structures and invariant densities
Letters in Mathematical Physics, vol. 106, pp. 1415-1427, 2016
27. Leapfrog map and Toda lattice, appendix to Integrable cluster dynamics of directed networks and pentagram maps by M. Gekhtman, M. Shapiro, S. Tabachnikov, and A. Vainshtein
Advances in Mathematics, vol. 300, pp. 390-450, 2016
28. Algebraic geometry and stability for integrable systems
Physica D, vol. 291, pp. 74-82, 2015
29. Singularities of bi-Hamiltonian systems, with A. Bolsinov
Communications in Mathematical Physics, vol. 331, pp. 507-543, 2014
30. Stability of relative equilibria of multidimensional rigid body
Nonlinearity, vol. 27, pp. 1419-1444, 2014
31. The derived algebra of a stabilizer, families of coadjoint orbits, and sheets
Journal of Lie Theory, vol. 24, pp. 705-714, 2014
32. Curvature of Poisson pencils in dimension three
Differential Geometry and its Applications, vol. 31, pp. 557-567, 2013
33. Stability of stationary rotations of multidimensional rigid body
Moscow University Mathematics Bulletin, vol. 68, pp. 80-82, 2013
34. Stability in bihamiltonian systems and multidimensional rigid body

- Journal of Geometry and Physics*, vol. 62, pp. 2414-2423, 2012
35. A note on relative equilibria of a free multidimensional rigid body
Journal of Physics A: Mathematical and Theoretical, vol. 45, p. 325203, 2012
 36. Algebra and topology of integrable systems: problems for investigation, with A. Bolsinov, A. Konyaev, and A. Oshemkov
Tr. Sem. Vektor Tenzor Anal., vol. 28, pp. 119-191, 2012 (in Russian)
 37. Classification of almost toric singularities of Lagrangian foliations
Sbornik: Mathematics, vol. 202, pp. 1021-1042, 2011
 38. Smooth invariants of focus-focus singularities
Moscow University Mathematics Bulletin, vol. 66, pp. 178-180, 2011